

INGA, Russian Federation

WMO: 307030

Lat: 52.971N

Lon: 101.959E

Elev: 543

StdP: 94.97

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-39.1	-36.3	-41.5	0.1	-38.9	-38.8	0.1	-36.2	7.8	-12.1	5.7	-13.2	0.1	230	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	27.9	18.2	26.3	17.6	24.7	16.7	19.9	25.0	19.0	24.4	18.1	23.1	2.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	13.9	22.4	17.1	13.0	21.3	16.1	12.3	20.4	59.7	25.0	56.6	24.5	53.6	23.4	23.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.4	5.0	4.1	-43.4	30.5	2.6	1.6	-45.3	31.6	-46.8	32.6	-48.3	33.4	-50.2	34.6		
(5)			-43.4	21.4	2.6	0.8	-45.3	21.9	-46.8	22.4	-48.3	22.8	-50.2	23.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.2	-21.8	-18.3	-8.5	1.7	7.7	14.8	17.1	15.0	7.9	-0.2	-11.4	-19.7	(6)
(7)		DBStd	14.65	6.02	6.84	6.58	4.94	3.97	3.59	2.40	3.19	3.97	5.08	7.08	7.25	(7)
(8)		HDD10.0	4673	985	792	574	251	94	4	0	4	89	316	643	921	(8)
(9)		HDD18.3	7167	1243	1025	833	499	329	115	52	111	314	574	893	1179	(9)
(10)		CDD10.0	577	0	0	0	2	24	148	221	158	24	0	0	0	(10)
(11)		CDD18.3	28	0	0	0	0	0	9	14	6	0	0	0	0	(11)
(12)		CDH23.3	639	0	0	0	2	59	248	208	112	11	0	0	0	(12)
(13)		CDH26.7	107	0	0	0	0	11	51	28	16	1	0	0	0	(13)
(14)	Wind	WSAvg	1.3	0.7	1.0	1.4	1.8	2.1	1.7	1.5	1.4	1.3	1.0	0.8	(14)	
(15)	Precipitation	PrecAvg	522	15	13	18	27	40	83	102	95	52	27	28	22	(15)
(16)		PrecMax	787	34	24	55	62	90	144	306	199	107	45	55	46	(16)
(17)		PrecMin	344	3	2	7	12	11	37	35	32	19	10	9	8	(17)
(18)		PrecStd	99	7	6	10	11	16	33	53	43	22	8	11	9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-4.0	1.3	10.9	22.2	28.8	30.4	29.5	28.5	24.7	17.8	6.5	-1.4	(19)
(20)			MCWB	-5.2	-2.5	4.3	9.7	14.6	18.3	19.2	19.5	14.3	8.4	2.1	-2.5	(20)
(21)		2%	DB	-6.8	-2.1	7.2	17.7	25.1	28.1	27.2	26.2	21.6	14.0	2.6	-4.8	(21)
(22)			MCWB	-7.7	-4.5	1.7	8.2	12.8	17.9	19.0	18.6	13.8	7.1	0.3	-5.7	(22)
(23)		5%	DB	-9.3	-4.6	4.7	14.6	22.2	26.2	25.7	24.2	19.0	10.8	0.4	-6.8	(23)
(24)			MCWB	-10.1	-6.3	0.4	6.3	11.1	17.1	18.6	17.7	12.2	5.4	-1.0	-7.5	(24)
(25)		10%	DB	-11.8	-7.1	2.0	11.4	19.1	24.4	24.2	22.3	16.0	7.8	-1.3	-9.1	(25)
(26)			MCWB	-12.4	-8.5	-1.2	4.9	9.4	16.0	17.8	16.9	11.0	4.2	-2.5	-9.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-5.0	-2.2	4.2	10.8	15.7	20.2	21.5	20.5	16.8	9.4	2.5	-2.6	(27)
(28)			MCDB	-4.2	0.3	10.1	19.6	27.1	26.9	25.8	25.8	22.7	16.9	5.3	-1.2	(28)
(29)		2%	WB	-7.8	-4.3	2.2	8.7	13.3	18.9	20.4	19.4	14.6	7.4	0.6	-5.6	(29)
(30)			MCDB	-6.8	-2.4	6.8	16.6	23.0	25.6	25.0	24.8	19.4	13.2	2.3	-4.9	(30)
(31)		5%	WB	-10.2	-6.2	0.6	6.7	11.6	18.0	19.4	18.4	13.0	5.9	-0.8	-7.5	(31)
(32)			MCDB	-9.4	-4.6	4.1	13.5	20.9	24.9	24.1	23.3	18.0	10.2	0.4	-6.8	(32)
(33)		10%	WB	-12.4	-8.5	-1.0	5.0	10.0	16.9	18.5	17.2	11.2	4.2	-2.5	-9.6	(33)
(34)			MCDB	-11.8	-7.2	1.8	10.7	17.9	23.1	23.0	21.5	15.6	7.3	-1.4	-9.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	14.2	17.4	16.9	13.9	15.5	14.4	11.8	11.9	13.5	13.0	12.5	12.3	(35)
(36)			MCDBR	14.7	18.5	18.0	20.6	23.5	18.8	15.9	16.4	18.8	18.2	12.7	12.1	(36)
(37)			MCWBR	13.7	15.9	12.7	11.2	11.8	9.5	8.6	9.1	10.6	11.5	10.1	11.3	(37)
(38)		5% WB	MCDBR	14.5	17.8	15.9	19.1	21.5	16.6	13.3	14.7	16.9	17.2	12.2	12.0	(38)
(39)			MCWBR	13.6	15.3	11.2	10.9	11.3	9.0	7.8	8.6	10.0	11.1	10.1	11.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.270	0.300	0.341	0.386	0.431	0.423	0.419	0.389	0.343	0.321	0.297	0.272	(40)	
(41)		taud	2.125	2.107	2.122	2.193	2.105	2.245	2.295	2.361	2.440	2.390	2.211	2.092	(41)	
(42)		Ebn at Noon	695	788	831	842	818	830	826	833	836	773	656	603	(42)	
(43)		Edh at Noon	70	98	120	128	148	130	122	108	88	75	64	60	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.15	2.18	3.66	4.84	5.41	5.74	5.24	4.55	3.38	2.13	1.19	0.83	(44)	
(45)		RadStd	0.07	0.13	0.18	0.23	0.33	0.33	0.45	0.28	0.28	0.14	0.09	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page